

Richard Dedekind

Dirichlet prefaces and notices, II

Gesammelte mathematische Werke, vol. III, pp. 408–427

[Printed page 408]

LVI. Notice of P. Bachmann, *The Theory of Cyclotomy and its Relations to Number Theory*

[Literature section of the *Zeitschrift für Mathematik und Physik*, vol. 18, pp. 14–24, 1873.]

I comply all the more gladly with Hofrat Schlömilch's request that I discuss the present work more closely in the *Zeitschrift für Mathematik und Physik*, since I have greeted its appearance with sincere pleasure. A great and highly interesting domain is here, for the first time, made easily accessible to the mathematical learner by an appropriate delimitation and a lucid presentation, and this while presupposing only very moderate preliminary knowledge of algebra and number theory.

The theory of cyclotomy, that is, the theory of equations of the form $x^m = 1$, where m denotes a given positive whole number, had, although many interesting properties of the roots of unity x had already been known earlier, first been brought back to its real principles by Gauss; it thereby became the starting point for an entirely new science of immeasurable extent. The seventh section of the *Disquisitiones Arithmeticae*, published in 1801, which bears the title *De aequationibus circuli sectiones definientibus*, contains a purely algebraic method for reducing the above equation, whose solution by means of trigonometric functions had long been known, to other equations of lower, and indeed of lowest possible, degree. Here the concept of irreducibility appears for the first time (Art. 341), a concept which became decisive for the entire direction of later algebra. Although Gauss makes only slight use of it (Art. 346), I nevertheless do not doubt that this fundamental principle guided him also

[Printed page 409]

in the discovery of the details, and that he preferred the synthetic presentation only for the sake of brevity. This is suggested especially by the weighty words (Art. 365):

omnique rigore demonstrare possumus, has aequationes elevatas nullo modo nec evitari nec ad inferiores reduci posse, etsi limites hujus operis hanc demonstrationem hic tradere non patiantur, quod tamen monendum esse duximus, ne quis adhuc alias sectiones praeter eas quas theoria nostra suggerit, e.g. in 7, 11, 13, 19 etc. partes, ad constructiones geometricas perducere speret, tempusque inutiliter terat.

The truth of the assertion contained in these words is easy to prove in the present state of algebra, especially since the development and generalization of Gauss's

ideas by Abel and Galois. In fact, from the seed planted by Gauss there has arisen a science which, in order to awaken in a non-specialist at least a dim idea of its character, one might perhaps call the science of the algebraic kinship of numbers, or, to use an expression chosen by me, the science of the kinship of fields. It becomes apparent, namely, that the peculiar constitution of an equation, the possibility of reducing its solution to that of other equations, can be clearly recognized only when, besides its roots, one also considers infinitely many other numbers rationally derivable from one or more of them. The totality of these numbers is precisely what I call a field: a system of numbers which reproduce themselves again and again under the four simplest rational arithmetical operations. On this view, properties of an equation become properties of the corresponding field; relations between equations appear as kinship between fields. In particular, when two fields A , B mutually penetrate one another, there always arises again a field, their least common multiple, or more briefly their product AB , whose nature depends essentially on the kinship of the two fields.

Alongside this development of algebra proper, to which the theory of cyclotomy gave the first impulse, and hand in hand with it, number theory has taken a magnificent upswing.

[Printed page 410]

Gauss's algebraic investigations in the domain of cyclotomy already required some, though very elementary, auxiliary propositions from number theory; soon, however, it became clear that conversely cyclotomy had become an inexhaustible source from which ever new and significant gains for number theory flowed. One may say that almost all later advances which number theory made in the hands of Gauss, Jacobi, Dirichlet, Eisenstein, Kummer, and Kronecker either owe their very origin to cyclotomy, or, what in some cases was still more remarkable, entered into a previously unsuspected connection with cyclotomy. Among the latter advances are the investigations of Gauss and Dirichlet on the class number of quadratic forms; among the former is Gauss's extension of the concept of a whole number, whose later generalization led to Kummer's creation of ideal numbers.

It is therefore surprising that, despite the role just briefly described which cyclotomy plays in the history of modern mathematics, and despite the great fame which it enjoys before other creations of Gauss at least equally profound — one need only think of the regular heptadecagon —, it is surprising that no textbook has appeared in which cyclotomy is presented as a rounded whole, with full consideration of the details investigated partly by Gauss and partly by his successors. It is therefore a most welcome undertaking of the author to fill this conspicuous gap in our mathematical literature by the present work; and I am pleased to be able to add, for which his name is already sufficient guarantee, that he has carried out this undertaking in excellent fashion. For anyone who intends a deeper study of algebra and its relations to number theory, this work will provide the best guide, since without presupposing extensive prior knowledge it introduces him into the midst of a very rich body of material which has hitherto not been easily accessible, and in which one must be thoroughly oriented if one wishes to proceed to higher investigations.

I now permit myself in what follows to communicate a series of remarks occasioned by individual passages, or also by whole sections, of the work. When I emphasize individual points at which I mention or recommend an arrangement or presentation different from that followed by the author, this is by no means done

in order to censure him.

[Printed page 411]

Everyone who has occupied himself thoroughly with a definite subject — and for eighteen years I have turned to this part of mathematics with special predilection — forms certain viewpoints peculiar to himself, which he holds to be especially valuable, while they appear less important to another; and I admit that much here is purely a matter of taste. But I gladly use this occasion to make, from my long occupation with this subject, some communications concerning my views, and also some digressions into related matters, in the hope that they will be welcome to some readers. I let them follow here without internal connection, as they arose while reading the work, merely referring to the relevant passage.

Lecture 3, nos. 2 and 3, p. 13. The proof of the existence of primitive roots of unity would, I believe, gain in clarity and precision by a small rearrangement. Since $\varphi(d)$ is defined as the number of d th roots of unity which belong to the exponent d , then, before the contrary has been proved, $\varphi(d)$ appears as depending not only on d , but possibly also on n ; and the applicability of the summation theorem proved in the preceding lecture for determining $\varphi(d)$ remains subject to doubts, which are removed only subsequently by the remarks in no. 3. The investigation is probably simplest if the concept of primitive roots of unity is put first and $\varphi(d)$ is defined as the number of primitive d th roots of unity.

Lecture 4, p. 20. In this lecture the first concepts from the so-called theory of higher congruences are communicated. A somewhat more extensive presentation, also containing the most important propositions on prime functions, would prove very useful on many later occasions, especially for the 5th, 17th, and 18th lectures; but it would of course also require much more space. The proof of Schönemann's theorem (p. 26), assuming Fermat's theorem, can be shortened by

[Printed page 412]

the theory of the transformation of symmetric functions, which the author does after all presuppose as known in several places, for example in Lecture 5, no. 4.

Lecture 5, nos. 6 and 7. The communication of a special case of Kronecker's general irreducibility theorem leads me to point out here the real nerve of his proof, and also of Arndt's. This can be done with relative brevity if one assumes as known some general concepts concerning whole algebraic numbers and some propositions from the theory of ideals, which I have partly proved in the second edition of Dirichlet's *Lectures on Number Theory*, and partly communicated without proof in the *Göttingen Gelehrte Anzeigen* of 20 September 1871. Let μ denote a primitive m th root of unity; then everything depends on the number-theoretic content of the number $(1 - \mu)$. If $m = 1$, then $1 - \mu = 0$; but if m is divisible by a single prime number p , hence is a power of that prime, then, as is immediately evident,

$$p = \varepsilon(1 - \mu)^{\varphi(m)},$$

where ε denotes a unit and $\varphi(m)$ the familiar function of number theory. Finally, if m is divisible by two or more distinct primes $p, q, \dots$, then the number $(1 - \mu)$, since it divides all numbers of the form $(1 - \mu^n)$, where n denotes any positive whole number, must, by the preceding case, also divide $p, q, \dots$ and consequently be a unit; this can also be concluded directly from the equation whose roots are all primitive m th roots of unity. Now let a be a power of a prime number p , and let $m = ab$, where b is not divisible by p . It is then well known that one obtains all primitive m th roots of

unity, each once only, by multiplying each primitive root of the equation $x^a = 1$ by each primitive root of the equation $x^b = 1$. If α is a definite one of the former and β a definite one of the latter, then Kronecker's theorem, in the following somewhat sharpened form, reads:

If the fundamental number, or discriminant, $\Delta(\Omega)$ of a field Ω is not divisible by the prime number p , then the equation $f(x) = 0$, irreducible in Ω , which is satisfied by $x = \alpha\beta$, also has as roots all $\varphi(a)$ products $\alpha'\beta$ corresponding to all $\varphi(a)$ primitive roots α' of the equation $x^a = 1$.

[Printed page 413]

The proof rests on the following points. All roots μ of the equation

$$f(x) = \Pi(x - \mu) = 0$$

are in any case of the form $\mu = \alpha'\beta'$, where α' , β' denote primitive roots of unity respectively of degrees a , b . Whenever β' is identical with β , one has $(\beta - \mu) = \beta(1 - \alpha')$, and therefore materially, that is, apart from a unit factor, $= (1 - \alpha)$. If, on the other hand, β' is different from β , then $(\beta - \mu) = \beta(1 - \alpha'\beta'')$, where β'' denotes a root different from 1 of the equation $x^b = 1$; hence, by the preliminary remarks, $(\beta - \mu)$ is a unit. Consequently

$$f(\beta) = \Pi(\beta - \mu) = \varepsilon'(1 - \alpha)^n,$$

where ε' is a unit and n denotes the number of roots $\mu = \alpha'\beta'$ for which $\beta' = \beta$. Therefore $f(\beta)$ will be divisible by

$$p = \varepsilon(1 - \alpha)^{\varphi(a)}$$

always and only when $n = \varphi(a)$, that is, when all $\varphi(a)$ products $\alpha'\beta$ really are roots of the same equation $f(x) = 0$ irreducible in Ω . This divisibility of the number $f(\beta)$ by p can be proved from the assumption that the fundamental number $\Delta(\Omega)$ of the field Ω is not divisible by p as follows. First, this assumption is identical with the assertion that p is divisible by no square of a prime ideal of the field Ω ; and this is in turn equivalent to the assumption that there exist infinitely many powers

$$s = p^f, p^{2f}, p^{3f} \dots$$

of the prime number p for which every whole number ω of the field Ω satisfies the congruence

$$\omega^s \equiv \omega \pmod{p}.$$

Since the coefficients of the function $f(x)$ are such whole numbers ω , it follows that

$$f(\beta)^s \equiv f(\beta^s) \pmod{p}.$$

Choose further, as is always possible, the power s of the prime number p so that $s \equiv 1 \pmod{b}$, hence $\beta^s = \beta$, and at the same time $s \geq \varphi(a)$. Since $f(\beta)$ contains the factor $(1 - \alpha)$ at least once, $f(\beta)^s$ is divisible by p , and it follows that $f(\beta) \equiv 0 \pmod{p}$ as well, as was to be proved.

By repeated application of this result one obtains evidently the following theorem, which is only slightly more general than Kronecker's:

[Printed page 414]

If m is relatively prime to the fundamental number $\Delta(\Omega)$ of the field Ω , then the equation whose roots are all the primitive m th roots of unity is irreducible in Ω .

Lecture 6, p. 43. In this lecture the proper theory of cyclotomy begins; its essence consists in reducing the equation

$$X = \frac{x^p - 1}{x - 1} = 0,$$

where p denotes a prime number, to equations of lower degree. I permit myself here some general remarks on the distribution of the material and on the method of the investigation.

Gauss's presentation falls into two essentially different parts. The first (Arts. 342–358) contains the successive decomposition of the system of all $(p - 1)$ roots r^k of the above equation into so-called periods, and the setting up of the equations corresponding to them; the second part treats the reduction of these equations to pure equations, that is, their solution by radicals. I consider this sharp separation extremely appropriate. The first part deals only with the numbers rationally expressible by r , which form a field R of degree $(p - 1)$; and it is, even if this does not appear in Gauss's presentation, essentially independent of the existence of primitive congruence roots. The second part, by contrast, requires the consideration of the latter, and also the introduction of an auxiliary field S of degree $\varphi(p - 1)$, which is formed from the roots of the equation $x^{p-1} = 1$. The real object of the investigation is the product RS of the two fields R and S , which at the same time is the field of degree $(p - 1)\varphi(p - 1)$ formed from the roots of the equation $x^{p(p-1)} = 1$. This essential difference between the two parts just named would make it seem to me more advantageous if the author had let the chief content of Lectures 9, 15, 16, 17, 18 follow directly after the seventh.

As to the method of development, it cannot be denied that in Gauss's synthetic presentation the striving for brevity has triumphed over the requirement of deriving everything from a single algebraic idea. During my first thorough study of cyclotomy in the Whitsun vacation of 1855, although I understood the details well, I nevertheless had

[Printed page 415]

to struggle for a long time before I recognized in irreducibility the principle to which I needed only to put simple, natural questions in order to be driven with necessity to all the details. After these thoughts were completed through a detailed occupation with the algebraic investigations of Abel and especially of Galois, and after the discovery, achieved at the beginning of December of the same year, of the most general relations between any two irreducible equations had brought them to a certain conclusion, I later followed the method then gained in my two winter lectures on cyclotomy and higher algebra in 1856–1858; and I still believe today that it is also appropriate for the learner. Instead of proceeding directly from the periods, as the author does with Gauss, one should ask for the number of the distinct conjugate values which a number $\alpha = F(r)$ contained in R , that is, rationally expressible through r , possesses, and which arise when r is replaced by all $(p - 1)$ roots r^k of the irreducible equation $X = 0$. If h denotes all those f incongruent numbers (mod p) for which $F(r^h) = F(r)$, then irreducibility immediately implies that these numbers h reproduce themselves by multiplication; and from this it follows further that they form all the roots of the congruence $h^f \equiv 1 \pmod{p}$, that

f is a divisor of $(p-1) = ef$, and that e is the number of actually distinct values $F(r^k)$, each of which repeats f times. Namely, all $(p-1)$ exponents k split into e complexes of f exponents hn each, to each of which the one value $F(r^n)$ corresponds. If one asks for the real existence of such e -valued numbers $\alpha = F(r)$, then the condition $F(r) = F(r^h)$ implies that the rational coefficients or coordinates a in the expression $\alpha = F(r) = \sum ar^k$, into which every number contained in R can always, and by irreducibility only in one way, be brought, must be groupwise equal in sets of f ; namely, that

$$\alpha = a\eta + a_1\eta_1 + \cdots + a_{e-1}\eta_{e-1}$$

must hold, where each of the e quantities $\eta, \eta_1, \dots, \eta_{e-1}$ denotes a so-called period, that is, a sum of f terms r^{hn} . If such a number α is actually to be e -valued, only further groupwise equalities among the e rational numbers $a, a_1, \dots, a_{e-1}$

[Printed page 416]

need be excluded; hence the periods themselves are such e -valued, and indeed conjugate, numbers. It follows at once that every e -valued number α is the root of an irreducible equation of degree e , and that every number $\beta = f(r)$ satisfying the conditions $f(r) = f(r^h)$, even without being e -valued, is rationally expressible through α . Thus all such numbers β form a field R_e of degree e , completely determined by e . The most general result, in which the whole first part of cyclotomy is contained and which is easily derived in the same way, is the following. If α is an e -valued number, α' an e' -valued number, and e'' is the greatest common divisor of e and $e' = d'e''$, then α' satisfies an equation irreducible in R_e of degree d' . The same can also be expressed as follows: the product of the fields R_e and $R_{e'}$ is the field $R_{e'''}$, where e''' denotes the least common multiple of e, e' ; and $R_{e''}$ is their greatest common divisor. This forms only a special case of the general law mentioned above, which can be expressed at least partially by the equation

$$(BC, A) = (C, AB)(B, A),$$

where A, B, C are any three fields, and the symbol (B, A) denotes the number of numbers contained in B which are independent with respect to A .

In the second part of cyclotomy the question is raised whether the irreducible equations considered in the first part are soluble by radicals, that is, reducible to pure equations. This question necessarily presses toward the introduction of the field S , and once the irreducibility of the equation $X = 0$ with respect to S has been recognized, it is immediately clear that the propositions of the first part can be transferred directly to the new domain. If one now asks for such numbers as satisfy pure equations, then one is driven in the same way to all the resolvents and to the use of primitive congruence roots, just as in the first part one was driven to the periods; and all the detail follows with the greatest ease.

The path described above is followed by the author only in part; but I believe that clarifying the algebraic principles in the classical and at the same time simplest example of cyclotomy is a very useful preparation for the deeper study of algebra.

[Printed page 417]

Lecture 9, no. 4. Kronecker's method for determining the sign $\varepsilon = \pm 1$ in the equation

$$\varepsilon \Pi(x^{2h-1} - x^{p-2h+1}) = \sum \left(\frac{k}{p} \right) x^k + (x^p - 1)f(x)$$

can be shortened by dividing by $(x-1)^{\frac{p-1}{2}}$ and then putting $x = 1$.

Lecture 14. After the reciprocity laws for cubic and biquadratic residues have been proved in the theories of complex numbers, it is of interest to return to the theory of rational numbers. Although this subject by its nature does not belong to the present work, I nevertheless permit myself to communicate here the following from a larger work on arbitrary cubic fields, whose completion and publication I am for the present prevented from undertaking by official duties. Let k denote a whole rational number whose cube root is irrational; then from the equation $x^3 = k$ there arises a pure cubic field whose fundamental number has the form $D = -3g^2$, where g is a whole number easily derived from k . If one now asks for all prime numbers p not dividing k , of the form $3n+1$, for which the given number k is a cubic residue, then by means of the reciprocity law one reaches the following interesting result, which was in essence already known to Gauss: all non-equivalent, primitive positive quadratic forms $ax^2 + bxy + cy^2$ in which $b^2 - 4ac = D$ split into three divisions with equally many individuals, the first of which forms a group. By the forms of this first division all and only those prime numbers p are represented for which k is a cubic residue. With its help, the determination of the number of ideal classes of the cubic field is reduced to a known part of the theory of theta functions.

Lectures 17 and 18. In founding the theory of ideal numbers in cyclotomy, the author follows the path taken by Kummer. The attempt to overcome some doubts about the rigor of Kummer's first presentation, later completed, first led me in the winter of 1855/56 to the theory of higher congruences, in particular to the decomposition of the function $(x^m - 1)$ into prime functions with respect to arbitrary prime moduli. These propositions, which incidentally, if I am not mistaken,

[Printed page 418]

were first published by Schönemann and have also been found in Gauss's Nachlass, allow, it seems to me, a simpler foundation of Kummer's ingenious theory. Retain the former meaning of X , and let q be a prime different from p which belongs (mod p) to the divisor f of $(p-1) = ef$. Then

$$X \equiv Q_0 Q_1 \cdots Q_{e-1} \pmod{q},$$

where the e functions Q denote prime functions of degree f . Let h again denote all the f numbers $1, q, q^2, \dots, q^{f-1}$ incongruent with respect to p , which form the roots of the congruence $h^f \equiv 1 \pmod{p}$, and let Q, Q' be any two of the e prime functions. Then the congruence

$$Q(y) \equiv 0 \pmod{q, Q'}$$

has each time f incongruent roots $y = x^{hn}$. If one now writes the e functions in the form

$$Q = x^f - ux^{f-1} + \cdots \pmod{q},$$

then it follows incidentally that

$$\Pi(x - \eta) = F(x) \equiv \Pi(x - u) \pmod{q},$$

where $F(x)$ denotes the function of degree e which vanishes for the e periods η of f terms. But this is by no means required for the further foundation. The e numbers which the author denotes by ψ can, in effect, be completely replaced by the e numbers which arise from the product

$$Q_0(r)Q_1(r) \cdots Q_{e-1}(r) = qf(r)$$

by omitting each time one of the e factors $Q(r)$. The proof, especially in the theorems on powers of the ideal prime numbers, is simplified if, as is always possible,

the e functions Q are chosen so that $f(r)$ is relatively prime to q , that is, so that $f(x)$ becomes a relative prime function to $X \pmod{q}$. Instead of this, however, one may also use the decomposition of the function X into e factors with respect to arbitrarily high powers of q . I note here that the author's definition (on p. 262) of the powers of ideal prime numbers leaves a small and easily removed doubt, insofar as it is not proved that a number which contains a prime factor m times must certainly contain it also $(m-1)$ times. Until this is done, it remains conceivable, for example, that a number contains a prime factor exactly six times and at the same time exactly eight times.

[Printed page 419]

A similar doubtful place is also found in the general theory of ideals which I presented in the second edition of Dirichlet's *Lectures on Number Theory*: the assertion that the concept of powers of a prime ideal is wholly independent of the accidental definition of the latter (§ 163, 4) is not properly justified, but it can easily be proved from the propositions immediately following it. The whole presentation can be considerably simplified by some rearrangements, and an even greater simplification will presumably succeed for those who approach the material without preconceptions and with fresh powers. Still, I believe that by this theory, whose construction cost me several years of indescribable exertion, I have achieved something useful. I am, to be sure, far from wanting to compare its value with that of Kummer's first creative idea; but apart from the aesthetic satisfaction afforded by the recognition that the same simple laws according to which, in rational number theory, numbers are composed from prime numbers recur in the theory of ideals of every field, the certainty of the general theory has the great practical advantage that in each special case one need not again and again seek out the first foundations, and that the discovery of the special laws is made extraordinarily easier. How great this benefit is may be seen quite clearly in the present case of cyclotomy: the nature of all ideals occurring here follows — as one will easily convince oneself — with a few strokes of the pen from the single remark that the p numbers $1, r, r^2, \dots, r^{p-1}$ are certainly incongruent with respect to every prime ideal not dividing p , because their differences all divide the number p .

I close my already too extensive remarks on the present work by once more most warmly recommending it for study to all who wish to acquire a thorough and many-sided knowledge of cyclotomy and its relations to number theory, and by expressing to the honored author my thanks for publishing these lectures of his.

Brunswick, 1 February 1873.

[Printed page 420]

Explanatory notes to the preceding article

Kronecker's original theorem on the irreducibility of the cyclotomic equations says that the equation $\varphi_n(x) = 0$ of the primitive n th roots of unity is irreducible in an algebraic field $K(a)$ if the equation discriminant of a is relatively prime to n . In the remarks on Lecture 5, however, Dedekind proves something sharper: that it already suffices if the field discriminant of $K(a)$ is relatively prime to n .

The investigations concerning the class number of cubic fields which Dedekind mentions in his remarks on Lecture 14 were later published by him in the paper “Über die Anzahl der Idealklassen in reinen kubischen Zahlkörpern” (XXIX); compare the introduction to that work.

Ore.

[Printed page 421]

LVII. Notice of the first edition of Riemann’s collected works

[*Göttingische gelehrte Anzeigen*, year 1876, pp. 961–965.]

If, in consequence of a request addressed to me, I permit myself to announce the appearance of this work in these pages, I do so not with the intention of entering more closely into its contents. For such a notice has already been composed, more perfectly than I could manage, by the actual editor, my highly honored friend Heinrich Weber in Königsberg, and will soon be published elsewhere, in the repertory of L. Königsberger and G. Zeuner. The purpose of the following lines is rather only to make some communications concerning the origin of the edition, which may perhaps be of some interest to the readers of the *G. G. A.*

Soon after Riemann’s death I received from Frau Professor Riemann the honorable commission to examine the scientific Nachlass and to publish what was suitable for publication. Three papers, namely those on trigonometric series, on the hypotheses of geometry, and the contribution to electrodynamics, I found completed in the cleanest handwriting, and I hastened to publish them. The rest, with few exceptions, was in an entirely unordered state, easily explained by Riemann’s many journeys; and before all else it was necessary to examine, according to their contents, the extraordinary number of separate leaves that had accumulated since Riemann’s student days, to recognize what belonged together, and to order it. Only someone who has looked into these papers can form a clear conception of the nature of this preparatory activity. For the most part they contain,

[Printed page 422]

in many repetitions, drafts for the papers published by Riemann himself and preparations for his lectures. In addition there are excerpts from the most varied works, copies of passages that had especially aroused Riemann’s interest, and drafts of letters. Many leaves are covered with formulas without any explanatory text, and all these things are sometimes united on one and the same folio sheet, written partly from top to bottom, partly in the opposite direction. Under these circumstances my efforts to order the papers met with only imperfect success; and although I had worked through Riemann’s works again and again with zeal, I could not conceal from myself that for deciphering the content of some papers I did not possess the full detail knowledge required, since my own studies as a whole were directed toward another domain of mathematics. I therefore at first contented myself with putting together in fair copies what was intelligible to me, and intended to proceed to publish some of it, especially the Paris prize memoir together with a commentary on its relations to the paper on the hypotheses of

geometry, when I was disturbed in my plan for several years by the necessary preliminary work for a second edition of Dirichlet's number theory. Soon after I had turned back to it, a new idea arose in Göttingen: to publish Riemann's works collected in full. Clebsch, Riemann's successor, probably stimulated by Wilhelm Weber, had seized on this idea with all his liveliness in the spring of 1872; and when he visited me at Whitsuntide I gladly entered into his plan to include the Nachlass as well in this edition. At my request Clebsch took over the chief direction of the undertaking and had all the papers sent to Göttingen in order to subject them to another inspection and examination, which I considered urgently necessary for the reasons given above. I myself, since soon afterwards a worrisome secondary office absorbing all my strength was entrusted to me for three years, could take part in the intended edition only to a small degree. It promised to proceed rapidly, and printing was soon to begin, when the sudden and deeply lamentable passing of Clebsch in November

[Printed page 423]

1872 brought the enterprise completely to a halt. After for some time I had received no news at all of its progress, I tried to persuade an outstanding mathematician and friend of Clebsch to take his place; unfortunately he was prevented by the most compelling reasons from fulfilling my request. Thus the matter again lay idle, since I myself, because of my duties, was unable to take up the edition. With Frau Professor Riemann's consent I finally resolved, in November 1874, to ask Professor H. Weber in Zürich, who by his works had proved himself one of the deepest connoisseurs of Riemann's creations, to take the great work entirely into his own hands; and the smaller my hope had been of seeing my request fulfilled, the greater was my joy when, despite serious reservations, he declared himself unconditionally ready to devote his powers wholly to this undertaking. From that moment the matter took the most gratifying and rapid course. Besides the careful revision of the papers already published, it was a matter of another exact examination of the entire Nachlass. This laborious work yielded, besides other successes whose enumeration here would lead too far, the very fortunate result that several components of the Nachlass whose meaning had escaped me, or whose value I had not fully appreciated, could be included in the edition or at least used for it. It is therefore solely Professor Weber's merit that the work could so soon, and in such completeness, be handed over to the mathematical public. My own collaboration, apart from a few trifles, was thereafter limited to participation in the revision of the proofs.

Brunswick.

[Printed page 424]

LVIII. Preface to the third edition of Dirichlet's *Lectures on Number Theory*. 1879

Concerning the origin of the present work, the first half of which essentially reproduces a lecture given by Dirichlet in Göttingen in the winter of 1856 to 1857, I have given the necessary report in the prefaces to the first two editions and in the Göttingen *Gelehrte Anzeigen* of 27 January 1864 and 20 September 1871. The

principal purpose of the edition, to introduce the beginner to the elements of number theory and especially to the theory of binary quadratic forms by transmitting as faithfully as possible Dirichlet's perfected lecture, has still been retained; and I have not been able to decide to make any substantial alteration in this part of the work. The additions that I had already inserted in the earlier editions for completion, which begin with the fourth supplement and are partly worked out from papers by Dirichlet and partly arose from my own investigations, I have again considerably increased. Especially to be mentioned is the broader presentation, contained in the final supplement, of the same ideal theory which I first published in the second edition, but in so compressed a form that the wish for a more detailed treatment was expressed to me from several sides. I have complied with this request all the more gladly since an investigation carried out jointly by my friend H. Weber in Königsberg and myself, and soon to appear, has yielded the result that the same principles can be successfully transferred to the theory of algebraic functions. The new presentation, in which much is borrowed verbatim from the work *Sur la théorie des nombres entiers algébriques* published three years ago,

[Printed page 425]

has of course required much more space; but this will probably find excuse if, as I hope, one is convinced that the unified character of the whole work has by no means suffered. I also use this opportunity to draw the reader's attention to the papers by Selling (in the tenth volume of Schlömilch's *Zeitschrift für Mathematik und Physik*, 1865) and by Zolotareff (in the sixth volume of the third series of the journal of Liouville and Resal, 1880), in which the theory of ideals is founded on that of higher congruences; and finally I may express the hope that Kronecker's relevant investigations, which stem from an earlier period, will soon be published.

Brunswick, 11 November 1880.

[Printed page 426]

LIX. Preface to the fourth edition of Dirichlet's *Lectures on Number Theory*. 1894

In the prefaces to the first three editions (1863, 1871, 1879–1880) and in the Göttingen *Gelehrte Anzeigen* of 27 January 1864 and 20 September 1871, I have made the necessary communications concerning the first origin and gradual expansion of this work; I refer to them here. Even now, in publishing the fourth edition, the first part, which essentially reproduces a lecture delivered by Dirichlet in Göttingen in the winter of 1856 to 1857 and extends to § 120, has remained almost unchanged; and I have contented myself with adding a few new remarks to my earlier notes in order occasionally to point out another method of proof or also more recent appearances in the literature. The same is true of my additions, which are partly worked out from papers by Dirichlet and partly arose from my own investigations. Only the final supplement, which treats the general theory of whole algebraic numbers, has undergone a complete revision; both the algebraic foundations and the properly number-theoretic foundations are presented in the greatest detail and from the point of view which, after many years of conviction, I consider the simplest, because it presupposes chiefly only a clear survey of the realm of

numbers and knowledge of the rational fundamental operations. The same conception also underlies some works on ideal theory which I hope to publish soon; and so it may be excused if I have treated some things more thoroughly than the immediate aims of the present work seem to require.

In the great Festschrift *Grundzüge einer arithmetischen Theorie der algebraischen Größen*, by which Kronecker in advance erected a beautiful monument to his teacher and friend Kummer, who passed away so soon after him, there are laid down the comprehensive ideas upon which another theory of ideals rises. By publishing it (1882 in Crelle's

[Printed page 427]

Journal, vol. 92), Kronecker fulfilled a wish which I had often expressed, most recently in June 1880 on the occasion of the unveiling of our Brunswick statue of Gauss, when it was also agreed that this paper should appear in Crelle's Journal before the *Theorie der algebraischen Funktionen einer Veränderlichen* worked out by H. Weber and me. Its content was also completely new to me, since on the basis of an old letter communication from 1857 I had believed that Kronecker's theory had to be sought along entirely different paths, which I indicated in § 10 of my work *Sur la théorie des nombres entiers algébriques* (1877). I do not yet consider it possible to pronounce a secure judgment on the advantages and disadvantages of this theory, whose great significance is beyond doubt, since in the paper only the fundamental ideas are sketched in broad strokes; I should therefore like to express the wish that one of Kronecker's numerous pupils undertake to work out a complete, systematic presentation of this theory. I recently tried to give a small contribution to this in the *Mitteilungen der Deutschen mathematischen Gesellschaft in Prag* (1892). If one considers what transformations other parts of mathematics, for example the theory of elliptic functions, have undergone over time since their first beginnings, one will find it very probable that still simpler foundations than those known so far will also be found for ideal theory. As such a foundation one may consider, for example, the theorem derived by me from ideal theory (pp. 465, 541, 577 of the second, third, and fourth editions of this work) concerning the greatest common divisor of two arbitrary whole algebraic numbers. I tried many years ago to take this path; but I did not succeed in obtaining an essential simplification, because I could give the direct proof of this theorem only with the same means that in essence underlie my theory of ideals as well. Nevertheless, I would recommend this path to the attention of younger mathematicians who enter this field of research without preconceptions, and to whom such a proof may therefore more easily succeed than it did to me.

Bad Harzburg, 30 September 1893.